Contextual Collective-Tube Model Predictive Control with Finite-Sample Safety Certificates

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Abstract

Data-driven stochastic model predictive control (MPC) typically calibrates uncertainty tubes from residual samples. Global tubes give simple finite-sample guarantees, yet may be conservative and uneven across heteroscedastic operating regimes. This paper develops contextual collective-tube MPC. The calibrated object is a complete finite-horizon residual trajectory, while the tube used online depends on a predictable context observed before planning. Offline, each local trajectory tube forms a chart together with its tightened constraints and terminal ingredients. A tube atlas collects these charts and specifies their certified escape risks and admissible transitions. Online, the controller selects a compatible chart, accounts for its certified risk, and solves the corresponding tightened tube-MPC problem. We prove transition-compatible recursive feasibility on certified containment events and give finite-sample escape-risk certificates for finite and nested atlases. We also provide a finite-mission continuation mechanism, conditional nominal initialization for recovery, and conservative bounded-shift corrections for deployment-law mismatch. Numerical experiments compare alternative tube constructions under common risk and confidence targets, illustrate mission-budget and recovery mechanisms, and quantify the effect of state/input-dependent deployment shift.

Keywords. model predictive control; stochastic MPC; tube MPC; conformal prediction; scenario optimization; recursive feasibility; finite-sample guarantees; feedback covariate shift.

1 Introduction

Model predictive control handles state and input constraints by repeatedly solving a finite-horizon optimal-control problem and applying the first input of the resulting plan. Its use in safety-critical applications relies on the explicit representation of constraints in the online optimization. With suitable terminal constraints and a terminal cost, standard formulations provide recursive feasibility, meaning that a feasible problem remains feasible at the next step, as well as stability of the nominal or robust closed loop (Mayne et al., 2000; Rawlings et al., 2017). For uncertain systems, tube MPC separates the optimized nominal trajectory from

the error between the nominal and actual states. An ancillary feedback law keeps this error within a prescribed tube, while the nominal constraints are tightened to accommodate the possible deviations (Mayne et al., 2005; Kouvaritakis and Cannon, 2016). Stochastic tube MPC extends this construction to probabilistic tubes obtained from distributional or sample information (Cannon et al., 2009a, 2011).

Learning-based control uses disturbance or prediction-error samples to calibrate safety margins when the underlying distribution is unknown. Scenario optimization provides finite-sample guarantees for randomized robust design and chance-constrained optimization (Calafiore and Campi, 2005; Campi and Garatti, 2008; Calafiore, 2010; Campi and Garatti, 2018), and has been used in stochastic MPC and randomized tube design (Calafiore and Fagiano, 2013b,a; Schildbach et al., 2014; Lorenzen et al., 2017; Hewing and Zeilinger, 2020). Conformal prediction offers a related route to distribution-free prediction sets under exchangeability, the condition that the joint sample law is unchanged by reordering its observations (Vovk et al., 2005; Shafer and Vovk, 2008; Lei et al., 2018; Angelopoulos and Bates, 2023). Applications to control include planning in dynamic environments and shrinking-horizon schemes designed to preserve recursive feasibility, as well as joint-in-time stochastic MPC for linear systems (Lindemann et al., 2023; Stamouli et al., 2024; Vogel et al., 2026).

A residual trajectory collects the prediction errors over the entire horizon. Calibrating a tube for this complete trajectory gives a single containment event that can be used directly in the deterministic tube-MPC argument. This approach appears in Vogel et al. (2026), who combine joint-in-time conformal regions with indirect-feedback MPC, in which the nominal state is propagated rather than reset to the measurement at every step. Another valid approach calibrates individual error margins and combines their risks through a union bound; its conservatism depends on the dependence between errors and on the tube geometry. A related collective-tube formulation of Calafiore (2026) calibrates a single reusable trajectory tube and couples its escape-risk certificate to measured-state tube MPC.

This paper extends single-tube trajectory certification to online selection among tubes adapted to regimes observed before planning. A global mixture certificate may conceal elevated escape risks within particular regimes, while independently calibrated local tubes may fail to support feasible switching. We couple simultaneous regime-wise risk bounds with verified tube transitions and a budget for a finite mission, defined as a prescribed number of MPC steps. Relative to Calafiore (2026), a key problem we address here is preserving calibration validity and feasible continuation as the selected tube changes. The statistical layer also augments deterministic switching schemes with certified risks for online tube selection. Switching among tubes has substantial precedent: Zhang et al. (2016) study feasibility and robustness for switched tubes with mode-dependent cross-sections, while Hernandez Vicente and Trodden (2019) analyze feasible switching among independently designed tube-MPC controllers. These deterministic constructions provide the background for coordinating different tube-MPC problems.

In the present paper, a *context* is a label from a finite set, observed before the current

optimization, that identifies a regime such as an exogenous load or environmental condition. We call this label *predictable* because it is available when the controller selects its tube. Calibration trajectories carrying the same label are grouped together to obtain a risk certificate conditional on that context. Each trajectory tube forms a *chart* together with its tightened constraints and terminal ingredients. Its *cross-sections* bound the error at individual prediction steps. A *tube atlas* collects these charts and a graph of admissible transitions between them. All charts share the nominal system matrices A, B and ancillary gain K, while their residual laws and tube geometries may differ.

Calibration validity requires a relation between the data used offline and the trajectories generated during deployment. In the *matched-law* setting, knowing the current context must determine the conditional distribution of the future residual trajectory: this distribution must agree with the calibration law for every control sequence that the MPC may select. An observed exogenous Markov regime provides one such setting. When residuals also depend on the state or input, the deployment law may instead differ from the calibration law. We then use a *bounded-shift* condition, which limits the increase in the probability of any residual event by a known factor. We give a concrete model satisfying this condition by allowing the weights of a disturbance mixture to depend on the state and input. The formulation states both alternatives explicitly.

The main technical issue is preserving recursive feasibility when the controller changes charts. The standard proof constructs a candidate plan by shifting the previous solution forward one step and appending a terminal-feedback move. Even if two tubes are individually valid, a change between them may invalidate this shifted plan. We therefore impose explicit *transition-compatibility* conditions. Let S_t^c be the set of errors allowed at prediction step t by chart c, and let H denote the prediction horizon. If chart c is followed by chart d, the edge $c \to d$ requires inclusions of the form

$$A_K^t S_1^c \oplus S_t^d \subseteq S_{t+1}^c, \qquad t = 0, \dots, H-1, \tag{1.1}$$

together with terminal-set and terminal-input inclusions. Here $A_K = A + BK$ is the error-dynamics matrix under ancillary feedback, and $\oplus$ denotes the Minkowski sum of sets. The inclusion ensures that the correction needed to initialize the next plan at the measured state, combined with the successor chart's error allowance, fits within the allowance of the preceding plan one step later. It therefore extends the usual tube-shift condition to changes between charts.

Contributions

Building on single-tube collective calibration (Calafiore, 2026), this paper couples context-wise risk certificates to verified chart transitions and finite-mission continuation. The binomial and order-statistic tools supply the statistical inputs to this atlas construction. A *compatibility-first* construction first enlarges the design-stage tube shapes to satisfy the transition inclusions and then calibrates a common scale. This order preserves compatibility while reducing unnecessary

enlargement. To support online operation, a *continuation reserve* sets aside enough risk allowance for the remaining mission. We also adapt the conditional nominal initialization of Hewing and Zeilinger (2018): when initialization at the measured state becomes infeasible, the preceding nominal prediction supplies a feasible backup. Its optimization remains feasible after arbitrary errors, while physical state safety retains a probabilistic guarantee. The experiments compare tube geometry under common risk and confidence targets and examine deployment shift. Throughout, trajectory-level calibration avoids marginal risk allocation within a horizon, while mission accounting adds the certified risks across MPC steps.

Organization

Sections 2 and 3 introduce the statistical setup and the conditions for a valid tube atlas. The finite-sample certificates, including the bounded-shift correction, are derived in Section 4. Sections 5 and 6 then present the controller and establish its guarantees, followed by the construction procedures and experiments in Sections 7 and 8. The scope of the results and the conclusions are discussed in Sections 9 and 10.

2 Preliminaries and problem formulation

2.1 Notation

For sets $A, B \subset \mathbb{R}^n$, $A \oplus B = \{a + b : a \in A, b \in B\}$ denotes the Minkowski sum and $A \ominus B = \{x : x + B \subseteq A\}$ denotes the Pontryagin difference. For a matrix L, $LA = \{La : a \in A\}$. The support function of a compact convex set S is $h_S(q) = \sup_{s \in S} q^\mathsf{T} s$. For a binomial random variable $Z \sim \mathrm{Bin}(N, p)$ with N trials and success probability p, define

$$B_N(m; p) = \mathbb{P}\{Z \le m\} = \sum_{i=0}^{m} \binom{N}{i} p^i (1-p)^{N-i}, \qquad m = 0, \dots, N, \tag{2.1}$$

with the convention $B_N(N; p) = 1$.

2.2 Plant, nominal model, and ancillary feedback

Consider the discrete-time system

$$x_{k+1} = Ax_k + Bu_k + w_k, \tag{2.2}$$

where $x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}^m$, and w_k is an additive model error. The admissible conditional laws of this error are specified below. The state and input are subject to constraints defined by compact convex sets:

$$x_k \in \mathcal{X}, \qquad u_k \in \mathcal{U}, \tag{2.3}$$

with $0 \in \operatorname{int} \mathcal{X}$ and $0 \in \operatorname{int} \mathcal{U}$. A stabilizing ancillary gain $K \in \mathbb{R}^{m \times n}$ is fixed, so the matrix

$$A_K = A + BK \tag{2.4}$$

has all its eigenvalues inside the unit circle and governs the error dynamics. In the certified operating mode, the nominal initial state is set equal to the current measurement; we refer to this as *measured-state initialization* or a *reset*. At time k, the optimizer computes a nominal trajectory $(z_{0|k}, \dots, z_{H|k})$ and nominal inputs $(v_{0|k}, \dots, v_{H-1|k})$ over H prediction steps, where $t|k$ denotes a prediction t steps ahead made at time k. These variables satisfy

$$z_{0|k} = x_k, \qquad z_{t+1|k} = Az_{t|k} + Bv_{t|k}, \qquad t = 0, \dots, H-1. \tag{2.5}$$

The applied input is

$$u_k = v_{0|k}. \tag{2.6}$$

If the plan were executed with ancillary feedback, the predicted future input would be

$$u_{t|k} = v_{t|k} + Ke_{t|k}, \tag{2.7}$$

where $e_{t|k}$ is the difference between the predicted actual state and the nominal state. Since the two states coincide at initialization, $e_{0|k} = 0$, and the prediction error evolves according to

$$e_{t+1|k} = A_K e_{t|k} + w_{t|k}, \qquad t = 0, \dots, H-1. \tag{2.8}$$

Collecting these errors over the horizon gives the residual trajectory

$$E_k = (e_{1|k}, \dots, e_{H|k}) \in \mathbb{R}^{nH}. \tag{2.9}$$

The conditional-law assumptions below allow dependence between the disturbances $w_{t|k}$ within a prediction horizon. Accordingly, each complete residual trajectory is treated as one calibration observation.

Remark 2.1 (Interpretation and acquisition of residual trajectories)**.** The trajectory E_k describes the errors that would arise if the selected nominal plan were continued with the ancillary feedback in (2.7). We call this hypothetical continuation the *prediction tail*; only its first error is necessarily realized before the controller replans. Calibration trajectories may come from independent simulator runs or reconstructed disturbance episodes, provided that they follow the reference distribution specified below. A simulator may differ from the nominal model when the conditional distribution of its residuals either matches deployment or satisfies the stated bounded-shift condition. The same requirement applies to linearization residuals and to data collected under another controller. ⋆

2.3 Operating contexts and statistical assumptions

Let $\mathcal{F}_k$ denote the information available before chart selection at time k, represented as a sigma-algebra. It includes the design and calibration data, together with the measured state and past closed-loop observations. These information sets form a filtration as time progresses. A context map γ, fixed before calibration, uses an observed feature ξ_k to assign a label from a finite set $\mathcal{G}$:

$$G_k = \gamma(\xi_k) \in \mathcal{G}, \qquad \xi_k \text{ is } \mathcal{F}_k\text{-measurable.} \tag{2.10}$$

The measurability condition means that the context is known when the chart is selected. We describe possible changes of context by a directed graph $\mathcal{E}_{\mathrm{G}} \subseteq \mathcal{G} \times \mathcal{G}$, which must include every transition that can occur:

$$(G_k, G_{k+1}) \in \mathcal{E}_{\mathrm{G}} \quad \text{almost surely.} \tag{2.11}$$

When every context-to-context transition must be treated as possible, the graph includes every ordered pair of contexts and is called complete. Tube construction uses a *design data set*, kept separate from the calibration data, to choose the tube shapes and the associated *trajectory scores*. Each score assigns a scalar value to a residual trajectory; its sublevel sets determine the candidate tubes. These design choices are fixed before the independent calibration data are used to compute certificates. Write the calibration sample and its context groups as

$$\mathcal{D}_{\mathrm{cal}} = \{(G_i, E_i) : i = 1, \dots, N\}, \qquad \mathcal{I}_g = \{i : G_i = g\}, \quad N_g = |\mathcal{I}_g|. \tag{2.12}$$

Assumption 2.2 (Context-wise calibration sampling)**.** Conditional on the design data and calibration labels, the trajectories $\{E_i : i \in \mathcal{I}_g\}$ are independent draws from a reference law Q_g^0. Deployment uses fresh random draws that are independent of the design and calibration data. The resulting residual trajectories may depend on those data through the selected controller, and their conditional laws must satisfy Assumption 2.3. For a *fixed-atlas certificate*, every candidate chart and its score function are chosen using only the design data and remain fixed during calibration. For a *nested-family certificate*, the tube family is fixed in advance, with its members growing as a radius increases. Calibration may then select the radius through a prescribed *order statistic*, namely a specified entry in the sorted list of calibration scores. $\star$

Assumption 2.3 (Uniform conditional deployment law)**.** Consider any history allowed by the model with current context $G_k = g$, and let π denote a nominal plan that the controller may select, continued with the ancillary feedback. Write $Q_{k,g}^{\pi,\star}(\cdot \mid \mathcal{F}_k)$ for the resulting conditional law of the residual trajectory. Uniformly over all such histories and plans, this law satisfies one of the following conditions:

$$Q_{k,g}^{\pi,\star}(\cdot \mid \mathcal{F}_k) = Q_g^0, \qquad \text{matched law,} \tag{2.13}$$

$$Q_{k,g}^{\pi,\star}(D \mid \mathcal{F}_k) \le \Gamma_g Q_g^0(D) \quad \text{for every measurable } D, \qquad \text{dominated law,} \tag{2.14}$$

where $\Gamma_g \geq 1$ is a known finite constant. In the dominated-law case, the probability of any residual event can increase by at most the factor Γ_g relative to calibration. For the matched case, we write $Q_g = Q_g^0$ and have $\Gamma_g = 1$. ⋆

Because the controller uses the calibration results, this conditional-law requirement includes the calibration information in $\mathcal{F}_k$. It therefore addresses changes in the residual distribution caused by data-dependent choices of the controller and of the states subsequently visited. Temporal dependence within each trajectory is allowed, whereas the calibration trajectories themselves must be independent under Assumption 2.2. Overlapping windows from a single run generally require a separate analysis that accounts for their dependence.

Remark 2.4 (Meaning of the deployment assumption)**.** Under (2.13), the context contains all the information needed to determine the future residual law, regardless of the remaining history or selected plan. The future states nevertheless depend on the future disturbances through the plant dynamics. A label identifying only the current state region will generally leave some state/input dependence in the disturbance law. Thus, applying this framework to errors caused by saturation or multiplicative uncertainty requires a justified bound of the form (2.14); expressing those errors as additive residuals does not by itself establish that bound. MPC methods tailored to multiplicative uncertainty address this modeling issue directly (Cannon et al., 2009b). Our second numerical example uses state/input-dependent mixture weights for which the required bound can be proved. ⋆

Remark 2.5 (A finite sufficient context)**.** Let G_k be an observed Markov chain whose transition probabilities are unaffected by the plant state or control inputs; this is what we mean by an *exogenous* regime. Suppose the disturbances satisfy

$$w_k = \mu_{G_k} + L_{G_k}\eta_k, \qquad \eta_k \text{ i.i.d. and independent of the chain.} \tag{2.15}$$

To obtain Q_g^0, start the chain in mode g, generate its modes and independent random draws η_k for H steps, and propagate (2.8) from zero. The resulting residual law is the same for every nominal plan, so (2.13) holds. The context continues to evolve within the prediction horizon, and persistent regimes with nonzero disturbance means can induce temporal correlation. For a continuous autoregressive disturbance, predicting the future law generally requires the current disturbance value; replacing that value by a finite context label would need additional justification. ⋆

Desired guarantees and locality. The confidence level $1 - \delta$ describes the probability, over calibration samples, that the reported risk bounds are valid. On that event, each bound controls the conditional probability that a residual trajectory leaves its tube. For a mission accepted by the initial feasibility and budget tests, Section 6 converts these bounds into state-safety and reset-feasibility guarantees with mission risk allowance ε. When reset initialization becomes infeasible, a recovery mode based on the preceding nominal prediction provides a

separate nominal-feasibility guarantee for every disturbance realization. State violations remain possible outside the certified containment event. The choice of context resolution also involves a tradeoff: finer labels can reduce variation within each regime, but leave fewer calibration samples N_g per label. Rare contexts therefore require more data or a justified merging rule, and every context used online must satisfy the conditional-law assumptions.

3 Tube charts and transition-compatible atlases

3.1 Trajectory tubes and cross-sections

A chart combines a trajectory tube with its MPC ingredients. Cross-sections bound the errors at individual prediction steps, and the terminal set constrains the nominal endpoint.

Definition 3.1 (Tube chart)**.** A tube chart is a tuple

$$c = (g(c), \mathcal{T}^c, S_0^c, S_1^c, \dots, S_H^c, \mathcal{Z}_f^c, \bar{r}_c), \tag{3.1}$$

where:

(i) $g(c) \in \mathcal{G}$ is the context of the chart;

(ii) $\mathcal{T}^c \subset \mathbb{R}^{nH}$ is a measurable trajectory tube;

(iii) $S_0^c = \{0\}$ and $S_t^c \subset \mathbb{R}^n$, $t = 1, \dots, H$, are compact convex cross-sections containing zero and satisfying

$$E = (e_1, \dots, e_H) \in \mathcal{T}^c \quad \Longrightarrow \quad e_t \in S_t^c, \qquad t = 1, \dots, H; \tag{3.2}$$

(iv) $\mathcal{Z}_f^c \subset \mathbb{R}^n$ is a compact convex nominal terminal set;

(v) $\bar{r}_c \in [0, 1]$ is a certified upper bound on the trajectory escape risk.

The escape event of chart c is

$$\mathsf{Esc}^c(E) = \{E \notin \mathcal{T}^c\}. \tag{3.3}$$

$\star$

The chart induces the tightened constraints

$$\mathcal{X}_t^c = \mathcal{X} \ominus S_t^c, \qquad \mathcal{U}_t^c = \mathcal{U} \ominus K S_t^c, \qquad t = 0, \dots, H, \tag{3.4}$$

where $\mathcal{U}_H^c$ is not used by the online problem. The terminal set must be compatible with the final state tightening:

$$\mathcal{Z}_f^c \subseteq \mathcal{X} \ominus S_H^c. \tag{3.5}$$

A convenient representation uses a scalar trajectory score $\phi_g(E)$ and a radius ρ to define

$$\mathcal{T}^{g,\rho} = \{E : \phi_g(E) \le \rho\}, \tag{3.6}$$

where ϕ_g may, for example, take the maximum normalized error over prediction times and the defining inequalities of a polytope. We call this a *collective* tube because a single score tests containment of the whole trajectory. Thus, even when the tube is described by many inequalities, calibration controls the single event $\phi_g(E) \le \rho$.

3.2 The tightened MPC problem for a fixed chart

For a fixed chart, we use the collective-tube MPC formulation of Calafiore (2026, Section IV, (15)) with chart-indexed tube and terminal data.

For a chart c and state x, define the feasible set $\mathcal{F}_H^c(x)$ as the set of nominal sequences $(z_0, \dots, z_H, v_0, \dots, v_{H-1})$ satisfying (2.5), $z_0 = x$, and

$$z_t \in \mathcal{X} \ominus S_t^c, \qquad t = 0, \dots, H-1, \tag{3.7a}$$
$$v_t \in \mathcal{U} \ominus K S_t^c, \qquad t = 0, \dots, H-1, \tag{3.7b}$$
$$z_H \in \mathcal{Z}_f^c. \tag{3.7c}$$

Given stage and terminal costs ℓ and V_f, the fixed-chart value is

$$V_H^c(x) = \inf_{(z,v)\in\mathcal{F}_H^c(x)} \left\{ \sum_{t=0}^{H-1} \ell(z_t, v_t) + V_f(z_H) \right\}. \tag{3.8}$$

The optimizer is denoted by $(z_t^{c,*}(x), v_t^{c,*}(x))$ when it exists.

3.3 Transition-compatible tube atlases

We organize the charts into an atlas whose directed graph specifies the changes allowed by the controller. This graph concerns transitions between complete MPC constructions, whereas $\mathcal{E}_{\mathrm{G}}$ describes changes of the observed context.

Definition 3.2 (Tube atlas)**.** A tube atlas is a pair $(\mathcal{C}, \mathcal{E}_{\mathrm{tr}})$, where $\mathcal{C}$ is a finite family of tube charts and $\mathcal{E}_{\mathrm{tr}} \subseteq \mathcal{C} \times \mathcal{C}$ is a directed edge set. For $g \in \mathcal{G}$, let

$$\mathcal{C}_g = \{c \in \mathcal{C} : g(c) = g\}. \tag{3.9}$$

If $(c, d) \in \mathcal{E}_{\mathrm{tr}}$, chart d is an admissible successor of chart c. ⋆

An edge can be stored once the following sufficient conditions have been verified. They ensure that the preceding solution can be shifted into a feasible plan for the successor chart.

Definition 3.3 (Transition-compatible edge)**.** Let $c, d \in \mathcal{C}$. Fix a terminal nominal feedback gain K_f and write $A_f = A + BK_f$ for its closed-loop matrix. The ordered pair $c \to d$ is transition-compatible if the following inclusions hold:

$$A_K^t S_1^c \oplus S_t^d \subseteq S_{t+1}^c, \qquad t = 0, \dots, H-1, \tag{3.10a}$$

$$A_f \mathcal{Z}_f^c \oplus A_K^H S_1^c \subseteq \mathcal{Z}_f^d, \tag{3.10b}$$

$$K_f \mathcal{Z}_f^c \oplus K A_K^{H-1} S_1^c \oplus K S_{H-1}^d \subseteq \mathcal{U}. \tag{3.10c}$$

$\star$

At $t = 0$, (3.10a) holds automatically because $S_0^d = \{0\}$; we retain this case to keep the indexing uniform. The remaining cross-section inclusions accommodate the error correction in the shifted plan. The terminal conditions ensure that its appended input and terminal state satisfy the successor chart's constraints.

Assumption 3.4 (Verified and viable atlas)**.** Every edge stored in $\mathcal{E}_{\mathrm{tr}}$ satisfies all the conditions in Definition 3.3. Moreover, for every $c \in \mathcal{C}$ and every next context g' allowed by $\mathcal{E}_{\mathrm{G}}$, at least one verified successor belongs to $\mathcal{C}_{g'}$. We refer to this successor property as *viability* of the atlas. Each context has at least one chart with a nonempty feasible domain, although admitting a particular mission also requires feasibility at its measured initial state and sufficient risk budget. $\star$

Remark 3.5 (The single-chart case)**.** With one context and one chart, (3.10a) reduces to the fixed-tube condition of Calafiore (2026, Definition 8): $A_K^t S_1 \oplus S_t \subseteq S_{t+1}$. Given a one-step disturbance set W, the standard reachable cross-sections $S_0 = \{0\}$ and $S_t = \bigoplus_{j=0}^{t-1} A_K^j W$ satisfy this relation with equality. Theorem 6.3 then becomes the usual shifted-tail argument for measured-state initialization. Classical invariant-tube MPC uses the related condition $A_K S \oplus W \subseteq S$, often allowing the nominal initial state to differ from the measurement (Mayne et al., 2005; Kouvaritakis and Cannon, 2016). Probabilistic reachable sets, which contain the prediction error with a prescribed probability, likewise need a compatible set recursion: a marginal probability bound alone cannot establish the required inclusions. The atlas extends these conditions across charts and associates them with simultaneous context-wise risk certificates. $\star$

Remark 3.6 (Fallback charts require verified incoming edges)**.** Enlarging a successor tube makes (3.10a) more demanding; in particular, $0 \in S_1^c$ requires $S_t^d \subseteq S_{t+1}^c$. A global *fallback chart*, intended for use when a local chart is unsuitable, must therefore be designed together with its incoming edges and satisfy the terminal tests. One possible construction uses a common sequence of reachable cross-sections, as in Remark 3.5, replicated and certified for each context. Connecting local charts to this fallback may require enlarging or removing some of them. Both transition compatibility and the desired risk bound must be verified for the resulting construction. $\star$

4 Finite-sample certificates for trajectory-tube escape risk

This section certifies the quantity

$$r_c = Q_{g(c)}\{E \notin \mathcal{T}^c\}, \tag{4.1}$$

where Q_g is the context-g residual-trajectory law. We first work in the matched-law setting. In the dominated-law setting, the same calculation uses the reference law Q_g^0, after which Section 4.3 converts the result into a deployment bound. In both cases, the event being calibrated is escape from the complete trajectory tube $\mathcal{T}^c$.

The independence of the design and calibration samples permits flexible tube design without invalidating the subsequent certificate. The context map and tube shapes may be chosen from design data, physical insight, or optimization. For a finite fixed atlas, the resulting chart list is kept unchanged during calibration. For a nested family, the score and rank-selection rule are fixed first, and the calibration scores then determine the tube radius.

The confidence level $1 - \delta$ concerns the random calibration sample: it lower-bounds the probability that all reported upper bounds are valid. We call this the *successful calibration event*. On this event, $\bar{r}_c$ bounds the probability that a fresh residual trajectory in the corresponding context leaves chart c. Since validity holds simultaneously across charts, the controller may select among them using the information available online, provided that the conditional-law requirement in Assumption 2.3 holds.

4.1 Finite-atlas binomial certificate

For a chart $c \in \mathcal{C}_g$, count the calibration trajectories that fall outside its tube:

$$M_c = \sum_{i \in \mathcal{I}_g} \mathbf{1}\{E_i \notin \mathcal{T}^c\}. \tag{4.2}$$

Allocate a confidence-failure allowance $\delta_c > 0$ to each chart, with $\sum_{c \in \mathcal{C}} \delta_c \leq \delta$. The one-sided upper binomial confidence bound is

$$U(N, m, \delta_c) = \begin{cases} 1, & N = 0 \text{ or } m = N, \\ \inf\left\{p \in [0,1] : B_N(m; p) \leq \delta_c\right\}, & 0 \leq m < N. \end{cases} \tag{4.3}$$

The certified risk is

$$\bar{r}_c = U(N_{g(c)}, M_c, \delta_c). \tag{4.4}$$

Proposition 4.1 (Simultaneous finite-atlas certificate)**.** *Assume that, conditionally on the design sample and on the context counts N_g, the calibration trajectories in each context are independent with common law Q_g. Then the certificate* (4.4) *satisfies*

$$\mathbb{P}\left\{r_c \leq \bar{r}_c \text{ for all } c \in \mathcal{C}\right\} \geq 1 - \delta. \tag{4.5}$$

Here the probability is taken over the calibration trajectories. Because the reference-law bounds hold simultaneously, they remain valid for a chart selected from the stored family; interpreting the selected bound conditionally at deployment also requires Assumption 2.3.

Proof. Fix a chart $c \in \mathcal{C}_g$ and condition on N_g. Under the stated sampling model, the miss indicators $\mathbf{1}\{E_i \notin \mathcal{T}^c\}$, $i \in \mathcal{I}_g$, are i.i.d. Bernoulli variables with success probability r_c. The one-sided Clopper–Pearson construction (Clopper and Pearson, 1934) gives

$$\mathbb{P}\{r_c > U(N_g, M_c, \delta_c)\} \le \delta_c. \tag{4.6}$$

Indeed, for any fixed p, the event $U(N, M, \delta_c) < p$ implies $B_N(M; p) \le \delta_c$, and the probability of this event under $M \sim \mathrm{Bin}(N, p)$ is at most δ_c by the definition of the binomial lower tail. A union bound over the finite chart family yields (4.5). The controller can subsequently choose charts adaptively, since the calibration event in (4.5) already covers every stored chart. □

4.2 Nested score/order-statistic certificate

When the chart radius is selected from calibration data, we instead use a family whose tubes are nested as the radius increases. Let $\{\mathcal{T}_g(\rho) : \rho \in \mathbb{R}_+\}$ satisfy $\mathcal{T}_g(\rho) \subseteq \mathcal{T}_g(\rho')$ whenever $\rho \le \rho'$. The associated *nonconformity score*, measuring how large a tube is needed to contain a trajectory, is

$$\phi_g(E) = \inf\{\rho : E \in \mathcal{T}_g(\rho)\}. \tag{4.7}$$

Sort the calibration scores in context g as $\phi_{g,(1)} \le \cdots \le \phi_{g,(N_g)}$. Before inspecting their values, choose $s_g \in \{0, \ldots, N_g - 1\}$, which specifies how many of the largest scores are skipped when selecting the radius. This choice may depend on N_g and on the risk and confidence targets. Set

$$\hat{\rho}_g = \phi_{g,(N_g - s_g)}. \tag{4.8}$$

The calibrated chart is $\mathcal{T}_g(\hat{\rho}_g)$. We assume that each family member has the sublevel representation $\mathcal{T}_g(\rho) = \{E : \phi_g(E) \le \rho\}$, which holds for the box constructions used below.

Proposition 4.2 (Order-statistic risk law)**.** *Assume that the scores in context g are i.i.d. with a continuous distribution. Let*

$$V_g = Q_g\{E \notin \mathcal{T}_g(\hat{\rho}_g)\} \tag{4.9}$$

be the conditional escape risk of the order-statistic tube. Then, for every $\varepsilon \in [0, 1]$,

$$\mathbb{P}\{V_g > \varepsilon\} = \sum_{j=0}^{s_g} \binom{N_g}{j} \varepsilon^j (1 - \varepsilon)^{N_g - j}. \tag{4.10}$$

For score distributions with atoms, the same expression is an upper bound when all trajectories tied at the selected threshold are included in the tube. Consequently, if ε_g is chosen so that the right-hand side of (4.10) *is at most δ_g, then*

$$\mathbb{P}\{V_g \le \varepsilon_g\} \ge 1 - \delta_g. \tag{4.11}$$

Proof. The continuous-score statement is the scalar case of Calafiore (2026, Remark 5 and Theorem 6), applied within context g with $N = N_g$ and boundary size $s = s_g + 1$. For atoms, independent continuous tie ranks give the randomized certificate; retaining all original threshold ties enlarges the tube and preserves its upper bound. □

Remark 4.3 (Scenario and conformal readings)**.** For $s_g = 0$, (4.10) is the sample-envelope law $(1-\varepsilon)^{N_g}$; for $s_g > 0$, it is the usual discarded-extremes certificate applied to complete residual trajectories. ⋆

4.3 Bounded deployment shift

We now convert a reference-law certificate into a deployment certificate under the dominated-law alternative in Assumption 2.3. This correction also covers feedback-induced changes in the residual distribution when the trajectory-law bound is uniform over the relevant histories and selected control plans. Likelihood ratios are likewise used in weighted conformal prediction and in prediction under feedback-induced distribution shift (Tibshirani et al., 2019; Fannjiang et al., 2022).

The domination condition requires absolute continuity of the deployment law with respect to the reference law: any event assigned zero reference probability must also have zero deployment probability. It further requires the Radon–Nikodym derivative, or likelihood ratio, to be bounded by Γ_g. When densities exist, this derivative is their pointwise ratio. Merely sharing the same support is weaker, since the ratio can still be unbounded. A bound estimated from finitely many observed ratios would therefore require its own statistical justification.

Proposition 4.4 (Risk inflation under bounded shift)**.** *Suppose* (2.14) *holds and the reference risks satisfy* $r_c^0 = Q^0_{g(c)}\{E \notin \mathcal{T}^c\} \le \bar{r}_c^0$ *simultaneously. On this calibration event, any chart and nominal tail chosen from the information in* $\mathcal{F}_k$ *satisfy*

$$\mathbb{P}\{E_k \notin \mathcal{T}^{c_k} \mid \mathcal{F}_k\} \le \bar{r}_{c_k}, \qquad \bar{r}_c = \min\{1, \Gamma_{g(c)}\bar{r}_c^0\}. \tag{4.12}$$

Proof. For $D_c = \{E : E \notin \mathcal{T}^c\}$, (2.14) gives $Q^{\pi,\star}_{k,g}(D_c \mid \mathcal{F}_k) \le \Gamma_g Q^0_g(D_c) \le \Gamma_g \bar{r}_c^0$. The probability is also at most one. □

Proposition 4.5 (Controlled-mixture sufficient condition)**.** *Suppose the context is an exogenous Markov chain with the same transition law in calibration and deployment. In context* g*, let the reference disturbance density be a mixture of* J *fixed component densities* $f_{g,j}$*:*

$$f_g^0(w) = \sum_{j=1}^{J} a_{g,j}^0 f_{g,j}(w), \qquad a_{g,j}^0 > 0. \tag{4.13}$$

Conditional on the exogenous context path, the reference disturbances are independent, with density $f^0_{G_{k+t}}$ *at prediction step* t*. At each deployment step, retain these component densities but allow their mixture weights* $a_{g,j}(\mathcal{F}_k, u)$ *to depend on the state and input. Suppose* $a_{g,j}(\mathcal{F}_k, u) \le$

$\kappa a^0_{g,j}$ *for every history and input allowed by the model, and use fresh independent random draws to sample the resulting conditional distributions. Then* (2.14) *holds with* $\Gamma_g = \kappa^H$*, uniformly over the selected tails.*

Proof. Each conditional disturbance density is bounded above by κf^0_g. Applying the chain rule over the horizon gives a joint context-and-disturbance path density ratio of at most κ^H, because the identical context-transition factors cancel. This bound is preserved when intermediate variables are integrated out and when the disturbance sequence is mapped deterministically to the residual trajectory in (2.9). Since the one-step bound holds for every allowed history and input, it also covers feedback-dependent mixture weights. □

The distinction between a one-step mixture and a trajectory mixture matters. Changing a weight from 0.05 to 0.15 gives a factor of 3 when the component is drawn once from a mixture of two fixed *trajectory* laws. If the component is redrawn at every step, the direct horizon bound becomes 3^H. Benchmark B uses one evolving disturbance process across successive, overlapping prediction windows and derives the trajectory factor from its one-step conditional distributions.

5 Contextual collective-tube MPC

5.1 Chart selection with a continuation reserve

Fix a mission of $T \geq 1$ MPC steps and an initial risk budget $B_0 = \varepsilon \in [0, 1]$. Selecting a chart allocates part of this budget to its certified deployment risk, including any required shift correction. We call this allocation the chart's *risk charge*, and write B_k for the allowance remaining before step k. The candidates consistent with the observed context and the preceding chart are

$$\mathcal{A}_k = \begin{cases} \mathcal{C}_{G_0}, & k = 0, \\ \{c \in \mathcal{C}_{G_k} : c_{k-1} \to c\}, & k \geq 1. \end{cases} \tag{5.1}$$

Checking only $\bar{r}_c \leq B_k$ ensures that the current chart is affordable, but may leave too little budget for a successor. We therefore compute a *continuation reserve* on the finite graph:

$$\begin{aligned} R_0(c) &= 0, \\ R_\ell(c) &= \max_{g' : (g(c), g') \in \mathcal{E}_{\mathrm{G}}} \min_{d \in \mathcal{C}_{g'} : c \to d} \{\bar{r}_d + R_{\ell-1}(d)\}, \qquad \ell \geq 1, \end{aligned} \tag{5.2}$$

where an empty minimum is $+\infty$. The quantity $R_\ell(c)$ is the risk allowance set aside for the next ℓ steps after chart c. For each possible next context, the inner minimization chooses a successor with the smallest current charge plus its remaining reserve. The outer maximization then protects against any context transition allowed by the graph.

A mission is admitted only when at least one chart $c \in \mathcal{C}_{G_0}$ is feasible at the initial state and its charge plus reserve fit within the budget:

$$V_H^c(x_0) < \infty, \qquad \bar{r}_c + R_{T-1}(c) \le \varepsilon. \tag{5.3}$$

At each admitted step $k < T$, select

$$c_k \in \arg\min_{c \in \mathcal{A}_k} \{V_H^c(x_k) + \lambda_\Phi \Phi(c) : V_H^c(x_k) < \infty,\ \bar{r}_c + R_{T-k-1}(c) \le B_k\}, \tag{5.4}$$

where Φ is an optional, fixed penalty on tube size and $\lambda_\Phi \ge 0$ is its weight. After applying the first nominal input, update the available budget according to

$$B_{k+1} = B_k - \bar{r}_{c_k}. \tag{5.5}$$

Each charge allocates probability allowance before the disturbance is observed, so a contained trajectory leaves the charged amount spent. If every realized first-step error remains in its selected cross-section, Theorem 6.4 guarantees a successor that is both feasible and affordable. A mission that fails the initial budget test is rejected before any control move; its duration or risk specification must then be revised. Physical operation after rejection requires a separately specified policy. More generally, positive finite-sample charges limit how long a fixed budget can sustain certified operation.

5.2 Conditional nominal initialization

A sufficiently large disturbance may place the measured state outside the feasible domain of every chart initialized at that measurement. To preserve a feasible nominal optimization, we follow the conditional-initialization principle of Hewing and Zeilinger (2018). The controller enters *nominal recovery mode* at the first failure of (5.4) and uses the previous plan's one-step prediction as its nominal initial state. Thus,

$$y_k = \begin{cases} x_k, & \text{while (5.4) is feasible,} \\ z^*_{1|k-1}, & \text{in nominal recovery mode.} \end{cases} \tag{5.6}$$

During recovery, the optimizer uses the same nominal constraints with $z_{0|k} = y_k$ and a verified successor in the observed context, independently of the remaining risk allowance. Recovery is retained until the mission ends, and its steps receive no new certificate based on a zero initial error. Resuming certified operation requires a new mission-admission test. Propagating the nominal state without resetting it to the measurement is also the basis of indirect-feedback stochastic MPC (Hewing et al., 2020).

To keep the applied input within its hard constraint for every measured state, use

$$u_k = \Pi_{\mathcal{U}}(v^*_{0|k} + K(x_k - y_k)), \tag{5.7}$$

where $\Pi_{\mathcal{U}}$ denotes Euclidean projection onto $\mathcal{U}$. In reset mode, $y_k = x_k$ and the feasible nominal input already lies in $\mathcal{U}$, so this expression reduces to $u_k = v^*_{0|k}$. During recovery, the initial offset $a_k = x_k - y_k$ may be large. Continuing the ancillary feedback without projection would give the error $A_K^t a_k + e_{t|k}$; projection may modify it further. Proposition 6.5 therefore establishes feasibility of the nominal optimization, while the projection enforces the input bound. Certifying physical state safety during recovery would additionally require error sets that account for the initial offset and projection. The mission bound in Corollary 6.8 covers the first event that makes recovery necessary.

Implementation. Offline, choose the tube shapes using the design data, calibrate them, and verify all edge and terminal conditions. The retained graph is then used to evaluate the reserve recursion (5.2). Online operation begins with the admission test and follows (5.4) for as long as it is feasible. When initialization at the measured state fails, (5.6) carries the previous nominal prediction into recovery. Budget rejection and recovery entry are recorded separately, since they have different causes. These admission and continuation rules make the required feasibility checks explicit.

6 Closed-loop guarantees

6.1 Scope of the closed-loop statements

During certified reset operation, the controller applies only $u_k = v_{0|k}$ before replanning at time $k+1$. Thus, the complete residual trajectory $E_k = (e_{1|k}, \dots, e_{H|k})$ describes the hypothetical continuation of that plan under the ancillary feedback, whereas its first component $e_{1|k}$ is the realized one-step residual. Recursive feasibility at the next step requires only $e_{1|k} \in S_1^{c_k}$. Full containment, $E_k \in \mathcal{T}^{c_k}$, is stronger: it certifies the entire planned horizon and supplies a conservative event for mission accounting. A failure of this certificate means that the residual trajectory has left its tube; an actual state or input violation may still be avoided.

6.2 Safety over one prediction horizon

The next lemma expresses the tube-containment safety implication of Calafiore (2026, Theorem 10) for a selected chart.

Lemma 6.1 (Joint horizon safety under tube containment)**.** *Let chart c be feasible at state x, with nominal solution (z_t, v_t). If the residual trajectory $E = (e_1, \dots, e_H)$ satisfies $E \in \mathcal{T}^c$, and if the ancillary input* (2.7) *is used along the prediction, then*

$$z_t + e_t \in \mathcal{X}, \qquad v_t + K e_t \in \mathcal{U}, \qquad t = 0, \dots, H-1, \tag{6.1}$$

with $e_0 = 0$. Moreover $z_H + e_H \in \mathcal{X}$.

Proof. Containment gives $e_t \in S_t^c$. The conclusions follow from the Pontryagin tightenings in (3.7), including the terminal tightening (3.5). □

Corollary 6.2 (One-step closed-loop safety)**.** *If chart c_k is feasible at x_k and $E_k \in \mathcal{T}^{c_k}$, then the implemented input $u_k = v_{0|k}$ satisfies $u_k \in \mathcal{U}$ and the next state satisfies $x_{k+1} \in \mathcal{X}$. The statement uses only $S_0^{c_k} = \{0\}$ and $e_{1|k} \in S_1^{c_k}$.*

Proof. Apply Lemma 6.1 at $t = 0$ for the input and at $t = 1$ for the next state, using its terminal assertion when $H = 1$. □

On the successful calibration event $\{r_c \le \bar{r}_c \ \forall c\}$, a chart c_k selected before planning therefore has a conditional horizon-certificate failure probability of at most $\bar{r}_{c_k}$ in the matched-law setting. In the dominated-law setting of Assumption 2.3, the same interpretation uses the inflated risk bound.

6.3 Recursive feasibility under chart transitions

The following result extends the fixed-tube shift argument of Calafiore (2026, Theorem 11) to different current and successor charts. The cross-chart inclusions and the terminal continuation in (6.3c) are verified explicitly.

Theorem 6.3 (Recursive feasibility for a transition-compatible atlas)**.** *Suppose chart c is feasible at time k. Let $(z_0^*, \dots, z_H^*)$ and $(v_0^*, \dots, v_{H-1}^*)$ be an optimal nominal state and input sequence, and suppose the first realized residual satisfies*

$$e_1 = x_{k+1} - z_1^* \in S_1^c. \tag{6.2}$$

Let d be any chart such that $c \to d$ in the verified graph of Assumption 3.4 . Define the shifted candidate at time $k+1$ by

$$\tilde{z}_t = z_{t+1}^* + A_K^t e_1, \qquad t = 0, \dots, H-1, \tag{6.3a}$$

$$\tilde{v}_t = v_{t+1}^* + K A_K^t e_1, \qquad t = 0, \dots, H-2, \tag{6.3b}$$

$$\tilde{v}_{H-1} = K_f z_H^* + K A_K^{H-1} e_1, \tag{6.3c}$$

$$\tilde{z}_H = A_f z_H^* + A_K^H e_1. \tag{6.3d}$$

Then $(\tilde{z}_0, \dots, \tilde{z}_H, \tilde{v}_0, \dots, \tilde{v}_{H-1}) \in \mathcal{F}_H^d(x_{k+1})$. Therefore the tightened MPC problem for chart d is feasible at time $k+1$.

Proof. Substitution of (6.3) into the nominal dynamics gives $\tilde{z}_0 = x_{k+1}$ and $A\tilde{z}_t + B\tilde{v}_t = \tilde{z}_{t+1}$, including the appended terminal move.

We next verify state constraints. Since the previous plan was feasible, $z_{t+1}^* \in \mathcal{X} \ominus S_{t+1}^c$ for $t = 0, \dots, H-1$, where for $t = H-1$ this follows from $z_H^* \in \mathcal{Z}_f^c \subseteq \mathcal{X} \ominus S_H^c$. The edge condition (3.10a) and $e_1 \in S_1^c$ give

$$A_K^t e_1 + S_t^d \subseteq A_K^t S_1^c \oplus S_t^d \subseteq S_{t+1}^c.$$

Thus

$$\tilde{z}_t + S_t^d = z_{t+1}^* + A_K^t e_1 + S_t^d \subseteq z_{t+1}^* + S_{t+1}^c \subseteq \mathcal{X},$$

so $\tilde{z}_t \in \mathcal{X} \ominus S_t^d$ for $t = 0, \dots, H-1$.

For $t = 0, \dots, H-2$, previous feasibility gives $v_{t+1}^* \in \mathcal{U} \ominus KS_{t+1}^c$. Again using (3.10a),

$$KA_K^t e_1 + KS_t^d \subseteq KS_{t+1}^c,$$

which yields $\tilde{v}_t \in \mathcal{U} \ominus KS_t^d$. For the terminal input, (3.10c) gives

$$\tilde{v}_{H-1} + KS_{H-1}^d \subseteq K_f \mathcal{Z}_f^c \oplus KA_K^{H-1} S_1^c \oplus KS_{H-1}^d \subseteq \mathcal{U},$$

so $\tilde{v}_{H-1} \in \mathcal{U} \ominus KS_{H-1}^d$. Finally, (3.10b) gives $\tilde{z}_H \in \mathcal{Z}_f^d$. Hence the shifted sequence is feasible for chart d. □

Theorem 6.4 (Feasibility with a mission reserve)**.** *Suppose Assumption 3.4 holds and the mission passes* (5.3)*. If the context follows $\mathcal{E}_{\mathrm{G}}$ and each preceding first-step residual lies in its selected cross-section $S_1^{c_k}$, then* (5.4) *remains feasible at every $k < T$ and the budget stays nonnegative.*

Proof. The admission test establishes the claim at $k = 0$. After selecting $c = c_k$, the reserve recursion ensures that every possible next context has a verified successor d satisfying

$$\bar{r}_d + R_{T-k-2}(d) \le R_{T-k-1}(c) \le B_k - \bar{r}_c = B_{k+1}, \qquad k < T-1. \tag{6.4}$$

When the first-step error is contained, Theorem 6.3 makes this successor feasible at the next measured state. It therefore satisfies both the feasibility and budget tests in (5.4). Induction, together with (5.5), completes the proof. □

Proposition 6.5 (All-realization nominal backup feasibility)**.** *Suppose the atlas is verified and viable, its cross-sections contain zero, and a nominal plan is initially feasible. For any realized x_{k+1}, every verified successor d admits a feasible nominal plan initialized at $y_{k+1} = z_{1|k}^*$. The controller can therefore repeat this nominal recovery step for any disturbance sequence.*

Proof. Use the previous nominal tail without an error correction: $\check{z}_t = z_{t+1|k}^*$ and $\check{v}_t = v_{t+1|k}^*$, appending $\check{v}_{H-1} = K_f z_{H|k}^*$ and $\check{z}_H = A_f z_{H|k}^*$. Since $0 \in S_1^c$, the edge conditions imply

$$S_t^d \subseteq S_{t+1}^c, \qquad A_f \mathcal{Z}_f^c \subseteq \mathcal{Z}_f^d, \qquad K_f \mathcal{Z}_f^c \oplus KS_{H-1}^d \subseteq \mathcal{U}. \tag{6.5}$$

The uncorrected tail satisfies the nominal dynamics by construction, and these inclusions ensure that it also satisfies the successor's tightened and terminal constraints. This argument is independent of the measured state. Viability of the graph supplies a successor at each subsequent step, so the construction can be repeated. □

6.4 Mission-level risk accounting

The fixed-tube deployment bound in Calafiore (2026, Corollary 12) uses a uniform risk charge. Here the charges follow the selected charts and the continuation reserve; concentration is stated for observable first-step escapes. Let I_k equal one while an admitted mission is in certified reset operation, and zero for rejected missions or after recovery has begun. Since the operating mode is known before chart selection, I_k is $\mathcal{F}_k$-measurable. Define the horizon-escape indicator and risk charge, together with the first-step escape indicator, by

$$F_k = I_k \mathbf{1}\{E_k \notin \mathcal{T}^{c_k}\}, \qquad p_k = I_k \bar{r}_{c_k}, \tag{6.6}$$

$$Y_k = I_k \mathbf{1}\{x_{k+1} - z_{1|k} \notin S_1^{c_k}\}. \tag{6.7}$$

All these quantities are defined to be zero when $I_k = 0$. The indicator F_k records escape from the full trajectory tube, whereas $Y_k \le F_k$ records escape of the first-step residual from its cross-section and is observable at time $k+1$. Either type of escape may occur while the physical constraints remain satisfied.

Lemma 6.6 (Predictable validity)**.** *Under Assumptions 2.2 and 2.3, let $\mathcal{A}_\delta \in \mathcal{F}_0$ denote the event on which the simultaneous reference-law calibration certificate is valid. This event has probability at least $1-\delta$. With the correction* (4.12) *applied when necessary, it satisfies*

$$\mathbb{E}(F_k \mid \mathcal{F}_k, \mathcal{A}_\delta) \le p_k. \tag{6.8}$$

Proof. On the calibration event, every stored chart has a valid reference-risk bound. The uniform condition in Assumption 2.3 transfers that bound to the conditional deployment law of the selected chart and tail. Since $\mathcal{A}_\delta$ is already measurable with respect to $\mathcal{F}_k$, the bound also holds under the stated conditioning. Multiplying by I_k gives (6.8). □

Theorem 6.7 (Finite-mission risk accounting)**.** *Under Lemma 6.6, suppose the accumulated charges satisfy the following bound on every realized path:*

$$\sum_{k=0}^{T-1} p_k \le \varepsilon. \tag{6.9}$$

Then

$$\mathbb{E}\left[\sum_{k=0}^{T-1} F_k \,\middle|\, \mathcal{A}_\delta\right] \le \varepsilon, \tag{6.10}$$

$$\mathbb{P}\{\exists k < T : F_k = 1\} \le \min\{1, \varepsilon + \delta\}. \tag{6.11}$$

For every $a > 0$, the observable one-step indicators also satisfy

$$\mathbb{P}\left\{\sum_{k=0}^{T-1} Y_k \ge \varepsilon + a \,\middle|\, \mathcal{A}_\delta\right\} \le \exp\left(-\frac{a^2}{2(\varepsilon + a/3)}\right). \tag{6.12}$$

An unconditional version of this tail bound is obtained by adding at most δ to its right-hand side.

Proof. The tower property gives $\mathbb{E}(\sum_k F_k \mid \mathcal{A}_\delta) \le \mathbb{E}(\sum_k p_k \mid \mathcal{A}_\delta) \le \varepsilon$. Markov's inequality yields the conditional any-failure bound; adding $\mathbb{P}(\mathcal{A}_\delta^c) \le \delta$ gives (6.11), irrespective of horizon overlap.

For the concentration bound, set $q_k = \mathbb{E}(Y_k \mid \mathcal{F}_k, \mathcal{A}_\delta) \le p_k$. Each centered difference $D_k = Y_k - q_k$ has zero conditional mean and is measurable at time $k+1$, so these differences form a martingale-difference sequence. They satisfy $D_k \le 1$ and $\sum_k \mathbb{E}(D_k^2 \mid \mathcal{F}_k, \mathcal{A}_\delta) = \sum_k q_k(1-q_k) \le \varepsilon$. The event in (6.12) implies $\sum_k D_k \ge a$, so Freedman's inequality (Freedman, 1975) applies. □

Corollary 6.8 (Mission state safety and reset feasibility)**.** *Let* Adm *denote successful admission. Under Theorems 6.4 and 6.7,*

$$\mathbb{P}\{\mathsf{Adm} \text{ and a state violation or recovery entry during the mission}\} \le \min\{1, \varepsilon + \delta\}. \quad (6.13)$$

For each fixed calibration realization in the successful event and each admitted initial condition, the corresponding conditional deployment probability is at most ε. The implementation (5.7) *also guarantees $u_k \in \mathcal{U}$ for every realization.*

Proof. Up to the first occurrence of $Y_k = 1$, one-step safety and Theorem 6.4 preserve both the state constraints and a feasible reset choice. A state violation or recovery entry therefore requires a preceding first-step escape, so $Y_k \le F_k$ allows us to apply (6.11). Fixing a successful calibration realization gives the ε bound by the same expectation argument, while projection ensures input admissibility. □

Overlap and conditioning. Evaluating F_k generally requires disturbances beyond time $k+1$, so centering it does not necessarily produce a one-step martingale difference. For this reason, the concentration result uses the observable indicators Y_k, while the expectation and any-failure bounds apply to F_k as well. Collective calibration avoids marginal union bounds within each prediction horizon, but mission accounting still adds risks across MPC steps and may be conservative when horizons overlap. There is also a distinction between joint and conditional admission probabilities: (6.13) bounds the joint event of admission and failure. Conditioning only on admission can alter the calibration-failure probability because the admission decision itself uses the calibration data.

6.5 Value decrease on certified containment intervals

On intervals of certified reset operation, we apply the practical value-decrease argument of Calafiore (2026, Theorem 14) to transition-compatible successors. Let

$$J_k = V_H^{c_k}(x_k) \quad (6.14)$$

be the value of the selected chart.

Assumption 6.9 (Terminal cost and Lipschitz regularity)**.** The stage cost ℓ and terminal cost V_f are continuous, nonnegative, and satisfy the terminal decrease condition

$$V_f(A_f z) - V_f(z) \le -\ell(z, K_f z) \tag{6.15}$$

for all $z \in \mathcal{Z}_f^c$ and all charts c. On the compact union of feasible nominal domains, ℓ and V_f are Lipschitz with constants L_ℓ and L_f. ⋆

Proposition 6.10 (Practical value decrease)**.** *Suppose Assumption 6.9 holds. Suppose chart c_k is feasible at time k, $E_k \in \mathcal{T}^{c_k}$, and c_{k+1} is selected among transition-compatible successors. Then*

$$J_{k+1} - J_k \le -\ell(x_k, u_k) + \sigma(\|e_{1|k}\|) + \Delta_k, \tag{6.16}$$

Here $\widetilde{J}_{k+1}$ is the cost of the shifted candidate (6.3) *in the successor problem, and $\Delta_k = J_{k+1} - \widetilde{J}_{k+1} \le 0$ measures the improvement obtained by reoptimizing. With the norm used to define the Lipschitz constants, one may take*

$$\sigma(r) = L_\ell \sum_{t=0}^{H-1} \left(\|A_K^t\| + \|K A_K^t\| \right) r + L_f \|A_K^H\| r. \tag{6.17}$$

Proof. By Theorem 6.3, the shifted candidate is feasible, so $\Delta_k \le 0$. Write $e = e_{1|k}$. Relative to its zero-error tail, (6.3) adds stage perturbations $A_K^t e$ and $K A_K^t e$, and terminal perturbation $A_K^H e$. The terminal decrease condition (6.15) bounds the zero-error candidate cost by $J_k - \ell(x_k, u_k)$. Lipschitz continuity therefore gives

$$\widetilde{J}_{k+1} - J_k \le -\ell(x_k, u_k) + L_\ell \sum_{t=0}^{H-1} (\|A_K^t\| + \|K A_K^t\|)\|e\| + L_f \|A_K^H\| \|e\|.$$

Use (6.17) and $J_{k+1} = \widetilde{J}_{k+1} + \Delta_k$ to conclude. □

7 Computation and offline atlas design

7.1 Polytopic tubes and inclusion tests

Suppose each cross-section is represented by finitely many linear inequalities:

$$S_t^c = \{e : H_t^c e \le h_t^c\}. \tag{7.1}$$

A facet normal is the outward normal to a boundary face of a polytope. Applying the support-function construction of Calafiore (2026, Proposition 16) to a candidate edge $c \to d$, (3.10a) is equivalent to

$$h_{A_K^t S_1^c}(q) + h_{S_t^d}(q) \le h_{S_{t+1}^c}(q) \quad \text{for all facet normals } q \text{ of } S_{t+1}^c, \tag{7.2}$$

provided that S_{t+1}^c has an irredundant representation, with one defining inequality per facet. Each support value can be obtained by solving a small linear program, and the terminal inclusions can be checked in the same way.

7.2 Compatibility-first calibration

For box cross-sections, *closure* means enlarging the preliminary boxes until they satisfy the desired cross-chart inclusions. This can be done before or after calibration, and the order affects the final tube sizes. Using only the design data, fix a desired graph and positive preliminary half-width vectors a_t^c. Set $\widetilde{s}_0^c = 0$ and $\widetilde{s}_1^c = a_1^c$, then compute the remaining half-widths componentwise:

$$\widetilde{s}_{t+1}^c = \max\left\{a_{t+1}^c, \max_{d:c\to d}\left(|A_K^t|\widetilde{s}_1^c + \widetilde{s}_t^d\right)\right\}, \quad t = 1, \dots, H-1. \tag{7.3}$$

Here $|A_K^t|$ denotes entrywise absolute value. The box enclosing $A_K^t \widetilde{S}_1^c$ has half-widths $|A_K^t|\widetilde{s}_1^c$, so each stage of the recursion supplies the widths needed for the corresponding inclusions. After $H-1$ stages, all desired cross-section inclusions hold. The resulting boxes are the *compatible templates* whose common scale is calibrated next.

Proposition 7.1 (Common-scale calibration of compatible templates)**.** *Let design-stage boxes $\widetilde{S}_t^c = [-\widetilde{s}_t^c, \widetilde{s}_t^c]$ satisfy* (3.10a)*. For each chart, calibrate*

$$\psi_c(E) = \max_{t=1,\dots,H}\max_{j=1,\dots,n}\frac{|e_{t,j}|}{\widetilde{s}_{t,j}^c} \tag{7.4}$$

using Proposition 4.2, with a rank chosen in advance and a confidence-failure allowance δ_c such that $\sum_c \delta_c \le \delta$. Denote the calibrated threshold by $\hat{\rho}_c$ and its reference-risk bound by $\bar{r}_c^0$. Define

$$\hat{\rho} = \max_c \hat{\rho}_c, \qquad S_t^c = \hat{\rho}\widetilde{S}_t^c, \qquad \mathcal{T}^c = \prod_{t=1}^{H} S_t^c. \tag{7.5}$$

All desired cross-section inclusions are preserved, and simultaneously $Q_{g(c)}^0\{E \notin \mathcal{T}^c\} \le \bar{r}_c^0$ with confidence at least $1-\delta$.

Proof. For every desired edge,

$$A_K^t S_1^c \oplus S_t^d = \hat{\rho}(A_K^t \widetilde{S}_1^c \oplus \widetilde{S}_t^d) \subseteq \hat{\rho}\widetilde{S}_{t+1}^c = S_{t+1}^c.$$

A union bound over the prescribed chart calibrations gives simultaneous validity of the individual score tubes. Since $\hat{\rho} \ge \hat{\rho}_c$, each final tube contains $\{\psi_c \le \hat{\rho}_c\}$ and has no larger escape risk. This union bound also applies when several charts share the same calibration trajectories. □

When deployment differs from calibration, apply the risk correction in (4.12). The terminal conditions must still be checked after scaling because they involve the fixed physical constraints. The common scale preserves the cross-chart inclusions while allowing the templates to retain their context-dependent shapes; separate scales would require the edges to be verified again. The reverse order, calibrating first and then closing the boxes, also preserves risk validity because closure only enlarges the tubes. We retain that construction as an *ablation*, a comparison that isolates the effect of the construction order, in Section 8.

7.3 Terminal construction and implementation

For the two-dimensional examples, set $K_f = K$ and choose a nonsingular coordinate transformation D such that $F = D^{-1}A_K D$ satisfies $\|F\|_\infty < 1$. In these coordinates, take the common terminal set $\mathcal{Z}_f = \{z : |D^{-1}z| \le a_f \mathbf{1}\}$, where $a_f > 0$ is its scale and $\mathbf{1}$ denotes a vector of ones. Write $x_{\max}$ and $u_{\max}$ for the componentwise bounds defining the state and input boxes, and s_t^c for the half-widths of the calibrated error sections. Sufficient terminal checks are

$$\begin{aligned} a_f|F|\mathbf{1} + \max_c |D^{-1}A_K^H|s_1^c &\le a_f\mathbf{1}, \\ a_f|D|\mathbf{1} + \max_c s_H^c &\le x_{\max}, \\ a_f|KD|\mathbf{1} + \max_{c,d}(|KA_K^{H-1}|s_1^c + |K|s_{H-1}^d) &\le u_{\max}. \end{aligned} \tag{7.6}$$

In these examples, the real and imaginary parts of an eigenvector provide suitable coordinates. A general matrix with eigenvalues inside the unit circle may require another terminal-set representation. The implementation selects a_f only when a positive value satisfies all the tests, and rejects designs with an empty feasible domain.

The complete procedure fixes the context map and tube shapes using the design data, then applies the fixed-atlas or nested-family calibration rule. Terminal verification and graph pruning may discard charts without invalidating the simultaneous certificate. Refitting tubes repeatedly to the calibration data would require an additional validity argument. After edges have been grouped by successor context, the reserve recursion has complexity $O(T|\mathcal{E}_{\mathrm{tr}}|)$. Online computation consists of deterministic tightened MPC problems for the admissible charts, preserving the standard tube-MPC optimization structure.

8 Numerical experiments

We next examine two benchmark examples: a double integrator in Section 8.2 and a mass-spring model in Section 8.3. All figures, data and tables in this section are reproducible via the companion code package available at `https://github.com/beppe969/collective-tube-mpc`.

8.1 Common protocol and baseline calibration

The stage cost is $\ell(x,u) = x^\mathsf{T}Qx + Ru^2$. Set $K_f = K$ and use $V_f(z) = z^\mathsf{T}Pz$, where P solves

$$A_K^\mathsf{T}PA_K - P = -(Q + K^\mathsf{T}RK). \tag{8.1}$$

Independent streams provide 1,800 design trajectories and 50,000 calibration trajectories per context. Separate tests use 80,000 trajectories per context. Each main comparison uses 120 missions of $T = 22$ steps, with paired initial states and primitive random streams. Initial modes have the stationary law of their exogenous chain. The common trajectory-risk target is $\beta = 0.004$ and the confidence allowance is $\delta = 0.001$ per method and experiment. The

mission budget is $B_0 = \varepsilon = 0.1$; the continuation-reserve constraint is enforced for every certified controller. A simultaneous confidence claim across all comparisons would require further allocation.

Preliminary half-widths are design-sample root-mean-square (RMS) values, bounded below by 10^{-8}. The *compatibility-first atlas* applies (7.3) before calibrating (7.4). For $G = |\mathcal{G}|$ contexts, the prescribed discard count and reference charge are

$$s = \max\{j : B_{N_g}(j; \beta) \le \delta/G\}, \qquad \bar{r}^0 = U(N_g, s, \delta/G). \tag{8.2}$$

The *calibrate-then-close ablation* first calibrates the raw RMS score and then enlarges its boxes by closure. The *no-edge diagnostic* retains those raw boxes. The *global comparator* uses the common shape given by the componentwise maximum of the context RMS shapes, then applies compatibility-first calibration under each conditional law and takes the largest threshold.

The *Bonferroni box* calibrates $d = nH$ absolute-coordinate margins at risk β/d and confidence $\delta/(Gd)$, and sums their certified risks. The *scenario box* contains the first N_s complete trajectories using their coordinatewise absolute maxima. This is the unique minimizer of the sum of d nonnegative half-widths subject to sample containment. Its dimension-dependent scenario bound is $B_{N_s}(d-1;\beta)$ (Calafiore, 2010); choose the smallest N_s making this at most δ/G. Both boxes undergo edge closure and terminal validation. These parameterizations represent specific baselines; alternative scenario-PRS constructions can have different effective dimensions.

Cost is the mean realized sum $\sum_{k<T} \ell(x_k, u_k)$. State tightening averages $s^c_{t,j}/x_{\max,j}$ over $t = 1, \dots, H$ and coordinates; input tightening averages $|K|s^c_t/u_{\max}$ over $t = 0, \dots, H-1$. Mission averages use the selected charts. Horizon escape is evaluated on ancillary continuations coupled to the actual first disturbance. The controlled kernel in Benchmark B is evaluated on each continuation's own state/input path. Physical violations and first-error escapes are logged separately.

The MPC problems are solved as convex quadratic programs (QPs), after eliminating the predicted state variables through the system dynamics (the standard condensed formulation). For the actual numerical solution, we used sequential least squares programming (SLSQP) with analytic derivatives and tolerance 10^{-10}. Acceptance requires successful termination and a primal residual below $2 \cdot 10^{-7}$; a separate linear program (LP) checks feasibility before a geometric rejection. The code also checks every stored edge and all terminal conditions in (7.6).

8.2 Benchmark A: a sufficient Markov context

Consider a double integrator with

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}, \quad K = \begin{bmatrix} -0.717 & -1.061 \end{bmatrix}, \qquad H = 8, \tag{8.3}$$

Table 1: Benchmark A: calibration and mission accounting, with $B_0 = 0.1$. Confidence is per context for scalar/scenario calibration and per absolute-coordinate margin for Bonferroni. Scalar shapes additionally use 1,800 independent design trajectories per context. The displayed spending and final balances hold in every main-run mission.

Construction	N used	Confidence	s	Chart charge	$\sum_k p_k$	B_T
Collective score	50,000	$\delta/3$	153	0.003993875	0.087865	0.012135
Bonferroni box	50,000	$\delta/48$	0	0.003448894	0.075876	0.024124
Scenario box	8,295	$\delta/3$	0	0.003999847	0.087997	0.012003

with $Q = \mathrm{diag}(1, 0.18)$, $R = 0.025$, $x_{\max} = (8, 4)^{\mathsf{T}}$, and $u_{\max} = 1.55$. Initial states are uniform on $[-6, 6] \times [-0.6, 0.6]$. The observed exogenous regime has transition matrix

$$P_{\mathrm{G}} = \begin{bmatrix} 0.970 & 0.025 & 0.005 \\ 0.250 & 0.730 & 0.020 \\ 0.150 & 0.100 & 0.750 \end{bmatrix}. \tag{8.4}$$

Use (2.15) with independent $\eta_k \sim \mathcal{N}(0, I_2)$ and

$$\begin{aligned} L_0 &= 0.25 \begin{bmatrix} 0.020 & 0 \\ 0 & 0.030 \end{bmatrix}, & \mu_0 &= (0, 0)^{\mathsf{T}}, \\ L_1 &= 0.25 \begin{bmatrix} 0.055 & 0.020 \\ 0.012 & 0.095 \end{bmatrix}, & \mu_1 &= 0.25(0.015, -0.010)^{\mathsf{T}}, \\ L_2 &= 0.25 \begin{bmatrix} 0.130 & 0.065 \\ -0.015 & 0.165 \end{bmatrix}, & \mu_2 &= 0.25(-0.020, 0.015)^{\mathsf{T}}. \end{aligned} \tag{8.5}$$

The context evolves throughout every prediction horizon. Its current value determines the future residual law independently of the selected tail, establishing the matched-law assumption exactly.

Table 1 specifies the common (β, δ) comparison. Bonferroni uses all 50,000 samples with zero discards, whereas the scenario box uses 8,295 samples and leaves the remainder unused. Thus the two coordinatewise maxima, and their resulting tightenings, differ. All three verified constructions have the same class of containment-based feasibility and mission-risk guarantee. The smallest calibration counts admitting these respective finite certificates are 1,998 for the scalar family, 8,295 for the dimension-16 scenario box, and 43,111 for the equal-allocation Bonferroni box. The scalar count excludes its independent design split; statistical admissibility still requires subsequent geometric checks.

Every row in Table 2 completes all missions without physical violations or reset failures. Compatibility-first calibration has seven horizon escapes and seven first-error escapes. Its independent conditional escape rates are 0.29875%, 0.29625%, and 0.27750%. The no-edge

Table 2: Benchmark A: 120 missions per row. Horizon is the escape percentage over 2,640 decisions; Test is the largest of three independent context-conditional escape percentages. Tightening and Feas. entries are percentages. Feas. is the stationary-context-weighted fraction of nominally feasible grid states. Every row except the no-edge diagnostic verifies all nine transitions and terminal conditions.

Method	Cost	Horizon	Test	X tight.	U tight.	Feas.
Compatibility-first	25.879	0.2652	0.2988	5.10	25.82	88.01
Calibrate then close	25.912	0.0000	0.0000	9.33	47.42	81.02
Global, compatible	25.929	0.0000	0.3425	5.68	29.44	86.81
Bonferroni atlas	25.887	0.0000	0.0037	7.01	35.50	86.83
Scenario-box atlas	25.884	0.0000	0.0325	6.46	32.60	87.38
No-edge diagnostic	25.890	0.1894	0.3387	2.16	11.30	–

diagnostic fails 89 scalar inclusion tests and verifies none of its nine candidate edges; sampled feasibility supplies no atlas-level recursive-feasibility certificate for that row. All budgets remain nonnegative. The calibrate-then-close and global constructions have the same charge and mission spending as the scalar row of Table 1. The no-edge row debits that reference charge diagnostically, without claiming the certified continuation property.

Compatibility-first calibration reduces both tightening measures by approximately 45% relative to calibration before closure. Its cost advantage over the scenario box is only about 0.02%. The main gain is geometric: LP tests on a 61×41 grid in $[-8, 8] \times [-4, 4]$ give the feasible-domain comparison in the table. These percentages describe the sampled nominal domain, rather than a probability of closed-loop success. Median QP times in these runs are approximately 1 ms, offering no evidence of a computational-speed advantage.

Data fairness and replication. Using the same total pool of 8,295 trajectories per context as the scenario box, the collective method reserves 1,800 for design and 6,495 for calibration. Its state/input tightenings are 5.41%/27.37%, compared with 6.46%/32.60% for the scenario box; costs are 25.880 and 25.884. Its charge is 0.003982643, giving spending 0.087618 and final balance 0.012382. This comparison accounts for the design-data requirement explicitly.

Three further independent design/calibration replications retain the geometric ranking against the implemented verified baselines. Across the four full-sample runs, collective state tightening ranges from 5.05% to 5.14% and input tightening from 25.54% to 26.02%. All 480 missions complete without physical violations or reset failures. One replication has an observed horizon-escape frequency of 0.5682%, above the 0.4% target; all independent conditional tests remain below it. Such finite empirical frequencies fluctuate, and overlapping horizon indicators provide no independent-binomial mission sample.

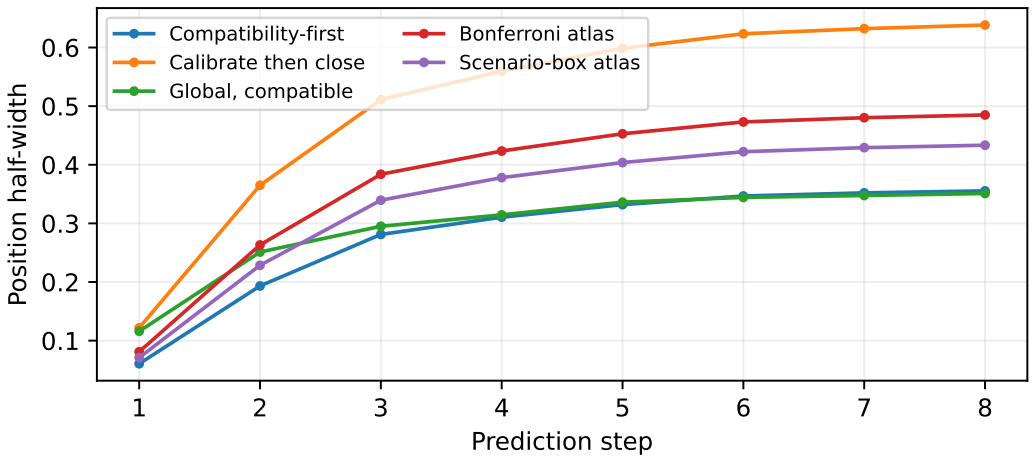


Figure 1: Benchmark A: position half-widths, averaged equally over contexts. Calibrating the compatible templates substantially reduces later-section enlargement.

Alternative text. Position half-widths over eight prediction steps for five verified constructions.

Why conditional calibration matters. A separate pooled-score diagnostic uses stationary initial-mode weights $(0.88394, 0.09109, 0.02497)$. Its mixture-weighted test escape is 0.3387%, while escape conditional on the rare third mode is 8.4013%. This is a mixture-law calibration diagnostic, with no chart-switching claim. The global comparator in Table 2 instead certifies every conditional law, avoiding that mismatch.

8.2.1 Budget continuation and nominal recovery

For a tighter mission budget, use two risk levels in each context:

$$(\beta_{g,h}, \beta_{g,l}) = (0.0040, 0.0012),\ (0.0037, 0.0015),\ (0.0031, 0.0018) \quad (g = 0, 1, 2). \tag{8.6}$$

The six preliminary templates multiply the RMS widths by their design-sample $(1 - \beta_{g,\cdot})$ score quantiles. Apply closure, allocate confidence $\delta/6$, and calibrate one common scale. All 36 edges and terminal conditions pass. Forty missions with $B_0 = 0.065$ and penalty $10^{-3} \sum_{t,j} s^c_{t,j}$ yield 12 different risk-level schedules. Spending lies between 0.062348 and 0.064881, with smallest final balance 0.000119. All missions complete without escapes or physical violations. This test uses the complete context graph and illustrates context-dependent charging with a continuation reserve.

For comparison, a current-affordability-only run using the single-rate atlas and $B_0 = 0.02$ exhausts its budget after five contained steps, spending 0.019969. The reserve admission test rejects that same 22-step specification before applying an input because it requires 0.087865. Budget exhaustion thus needs no realized failure.

A separate recovery stress test starts at $(1, 0.1)^\mathsf{T}$ and adds $(6.5, 3)^\mathsf{T}$ to the disturbance at $k = 3$. It is excluded from stochastic coverage estimates. At $k = 4$, the state is approximately $(6.4969, 2.9933)^\mathsf{T}$, inside the state box but outside the reset-feasible domain. Conditional initialization supplies 14 feasible recovery QPs through step 17. Projection preserves the input bound, while four subsequent states violate the state constraints. This separates current state admissibility from recoverable nominal feasibility. Statistical charging ends at recovery entry, after spending 0.015975.

8.3 Benchmark B: state/input-dependent deployment shift

We use a mass-spring model with

$$A = \begin{bmatrix} 1 & 0.35 \\ -0.2275 & 0.937 \end{bmatrix}, \quad B = \begin{bmatrix} 0.06125 \\ 0.35 \end{bmatrix}, \quad K = \begin{bmatrix} -1.371 & -1.713 \end{bmatrix}, \quad H = 10, \tag{8.7}$$

with $Q = \mathrm{diag}(1, 0.25)$, $R = 0.08$, $x_{\max} = (4.5, 3)^\mathsf{T}$, and $u_{\max} = 4$. Initial states are uniform on $[-3.2, 3.2] \times [-0.5, 0.5]$. The observed regime and innovation matrices are

$$P_{\mathrm{G}} = \begin{bmatrix} 0.96 & 0.04 \\ 0.18 & 0.82 \end{bmatrix}, \quad L_0 = 0.4 \begin{bmatrix} 0.012 & 0 \\ 0.002 & 0.018 \end{bmatrix}, \quad L_1 = 0.4 \begin{bmatrix} 0.030 & 0.010 \\ -0.004 & 0.042 \end{bmatrix}. \tag{8.8}$$

Each reference step independently chooses $L_g\eta$ or $3L_g\eta$, with $\eta \sim \mathcal{N}(0, I_2)$ and heavy-component probability $\alpha_0 = 0.05$. Deployment uses

$$\begin{aligned} \alpha_\star(x, u) &= \alpha_0[1 + (\Gamma^{1/H} - 1)h(x, u)], \\ h(x, u) &= \tanh\big((x_1/2)^2 + (x_2/1.5)^2 + (u/2)^2\big). \end{aligned} \tag{8.9}$$

The tested range $1 \le \Gamma \le 3^{10}$ gives valid probabilities. The heavy-weight ratio is at most $\Gamma^{1/H}$ and the light-weight ratio at most one. Proposition 4.5 therefore proves the trajectory factor Γ uniformly over every selected tail, including its state/input dependence.

At $\Gamma = 3$, the corrected atlas calibrates at $\beta/3$ with confidence $\delta/2$ per context. Its deployment charge is 0.003997602, giving $\sum_k p_k = 0.087947$ and $B_T = 0.012053$ in every mission. It completes all 120 missions without physical violations, reset failures, or budget exhaustion. State/input tightenings are 11.91%/28.17%, and mean cost is 10.969. The uninflated diagnostic has 9.61%/22.75% tightening and essentially the same cost. Its unadjusted spending is 0.087638; the justified shift-adjusted spending would be 0.262914, failing the 0.1 admission requirement. Consequently that row is diagnostic operation only.

The sweep in Table 3 uses the same reference samples and paired test streams. Independent tests execute nominal K-feedback tails from $[-2.4, 2.4] \times [-0.35, 0.35]$, using 80,000 trajectories per context. Their finite empirical rates concern those specified tails; uniform validity comes from the density-ratio proof. Additional accepted settings use 40 missions per variant. Every corrected mission completes without physical violations or reset failures.

Table 3: Benchmark B: shift-severity sweep. $N_{\min}$ is the minimum scalar calibration count per context; 50,000 are available. Tightening is for the corrected atlas, and test columns give the largest conditional escape-%. The last column is corrected mission spending with $B_0 = 0.1$; its difference from 0.1 gives B_T. Rejected designs have no corrected performance entry.

Γ	$100\alpha_{\max}$	$N_{\min}$	U tight.	Uninflated	Corrected	$\sum_k p_k$
1	5.00	1,897	22.82	0.3162	0.3162	0.087638
3	5.58	5,697	28.17	0.3312	0.0975	0.087947
10	6.29	18,999	37.10	0.3625	0.0087	0.083830
25	6.90	47,502	48.23	0.3887	0.0013	0.083604
100	7.92	190,019	–	0.4338	–	Reject
59,049	15.00	112,206,419	–	0.7388	–	Reject

The corrected designs at $\Gamma = 100$ and $\Gamma = 59{,}049$ are rejected for insufficient calibration data. At the largest shift, the uninflated context-conditional escape rates are 0.59625% and 0.73875%. Simultaneous two-sided 95% binomial intervals have lower endpoints 0.5369% and 0.6725%, both above the target. At that severity, the stated uniform correction requires over 112 million calibration trajectories per context. The example thus exhibits undercoverage when shift is ignored and the sample cost of a conservative trajectory-law bound. All accepted designs and rejections are reported, rather than selecting only favorable cases.

The numerical advantage in Benchmark A concerns the specified tube geometry and persists at matched total data. Cost differences remain small. All verified baselines carry the same class of containment-based guarantees; other score families can change their ranking. The operational tests document separate budget and recovery behavior, while the shift sweep exposes a limit of uniform likelihood-ratio protection.

9 Limitations and scope of the guarantees

The proposed approach uses a fixed linear nominal model and ancillary gain, with statistical guarantees based on independent calibration trajectories and an explicit relation between the conditional calibration and deployment laws. The context label must support either a matched law or a uniformly dominated law; its availability before planning is only the information requirement. Establishing a useful density-ratio bound can be demanding over long horizons, and the available calibration data may be insufficient for the requested certificate.

With verified chart transitions, containment of each first-step error preserves measured-state recursive feasibility. The continuation reserve additionally keeps an admitted mission affordable along those paths. After an arbitrary error, conditional initialization preserves nominal feasibility and projection enforces the hard input bound, while physical state safety remains subject to the mission-risk guarantee. Calibration confidence and deployment risk describe separate sources of uncertainty, and their mission-level composition remains additive

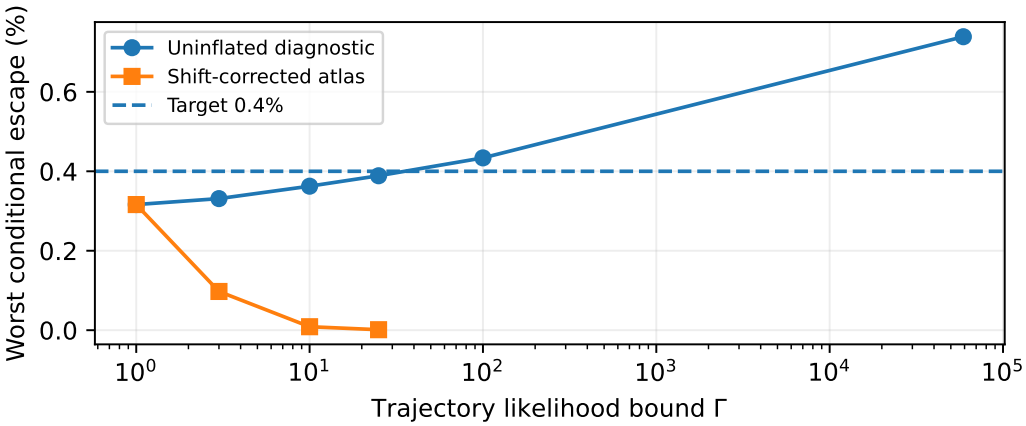


Figure 2: Benchmark B: worst conditional test escape versus trajectory shift bound. The last two corrected designs are unavailable with the prescribed calibration sample.

Alternative text. Worst context-conditional test escape versus the shift factor.

and potentially conservative for overlapping horizons.

10 Conclusion

Building on collective trajectory-tube calibration (Calafiore, 2026), this paper developed an atlas for predictable context changes. Its charts carry context-wise risk certificates, while verified transitions ensure that the shifted plan remains feasible in the successor chart. This coupling supports contextual tube selection with finite-mission continuation.

A further theme of the paper is that several sources of feasibility have to be distinguished. Transition compatibility deals with the geometric feasibility of successive MPC problems, whereas the continuation reserve ensures that enough certified risk remains available for the rest of a finite mission. When an unusually large disturbance makes measured-state initialization infeasible, conditional nominal initialization provides a feasible optimization backup, while physical state safety retains its probabilistic interpretation. The numerical examples also show that the order in which geometric compatibility and statistical calibration are imposed can have a significant effect on tube size, and that distribution shift can rapidly increase the amount of data required for a valid certificate.

These observations suggest several directions for further research. One is to learn or adapt the context partition itself while retaining finite-sample validity; this would require accounting explicitly for data-dependent context selection. A second direction is to replace independent calibration trajectories by dependent or closed-loop data, which would make the framework considerably easier to apply to long experimental records. At the mission level, the

present additive risk accounting is conservative because successive prediction horizons overlap. Exploiting this dependence could lead to substantially sharper long-horizon guarantees. It would also be useful to develop less conservative alternatives to uniform likelihood-ratio bounds for state- and input-dependent uncertainty.

Finally, the atlas viewpoint is not intrinsically tied to linear systems or to fixed ancillary feedback. Extending it to local or linear parameter-varying (LPV) models, nonlinear tube constructions, and controllers whose local feedback laws also vary with the context would broaden the range of applications. In that setting, an important question is how statistical certificates, geometric transition conditions, and online adaptation can be designed together so that changes in the local model or controller preserve a certified handover from one operating regime to the next.

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